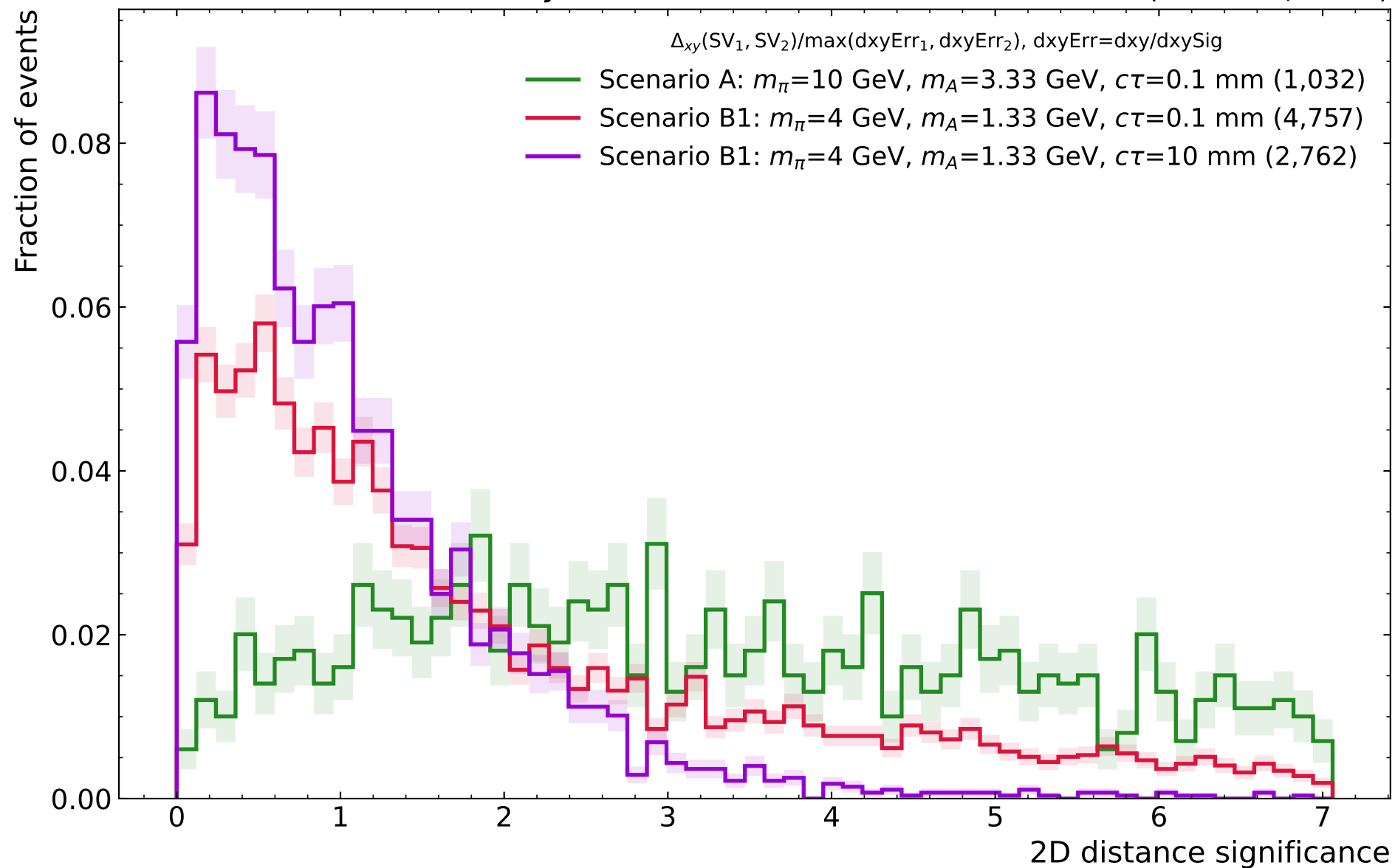


quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only

CMS *Simulation Preliminary*

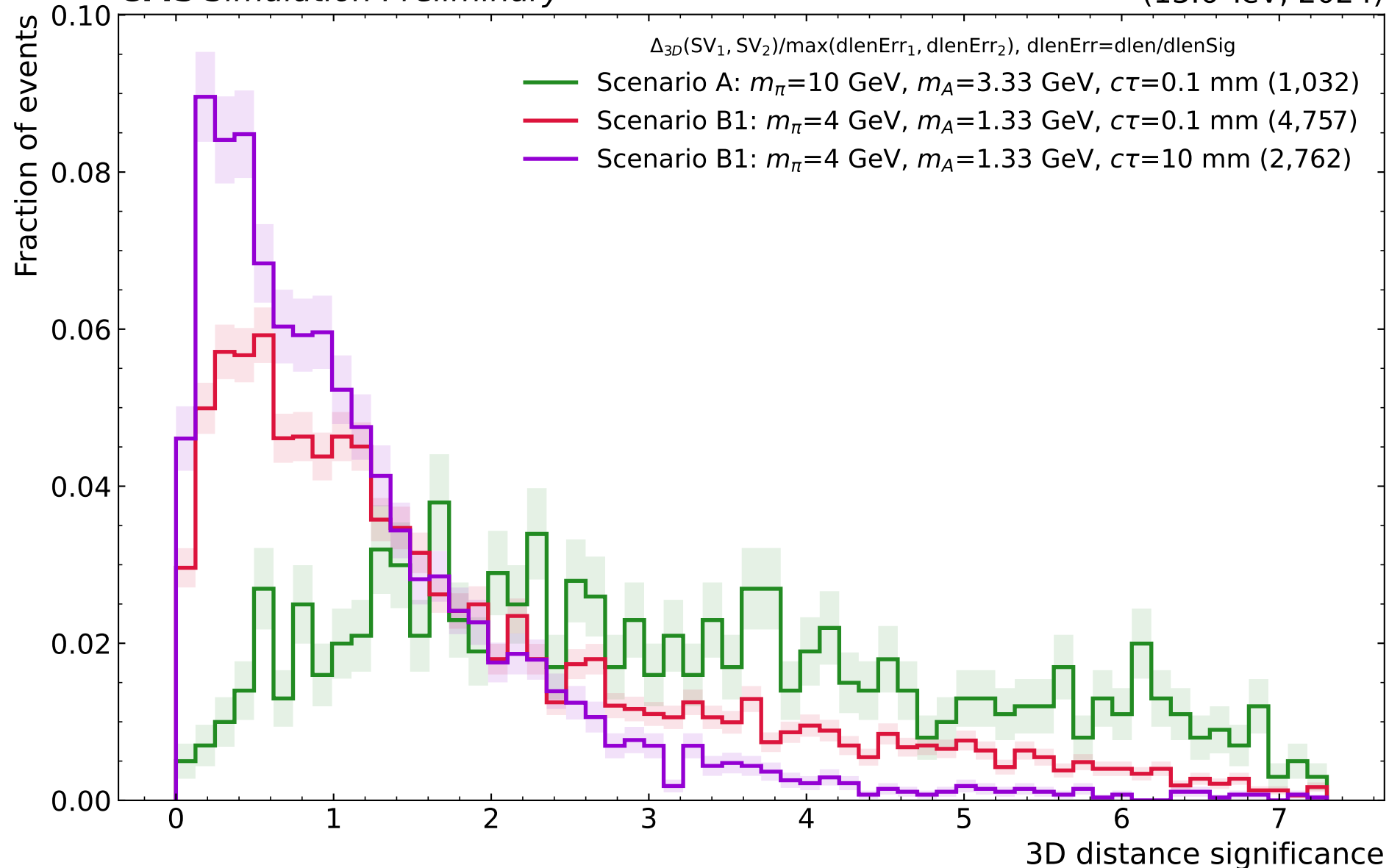
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only

CMS Simulation Preliminary

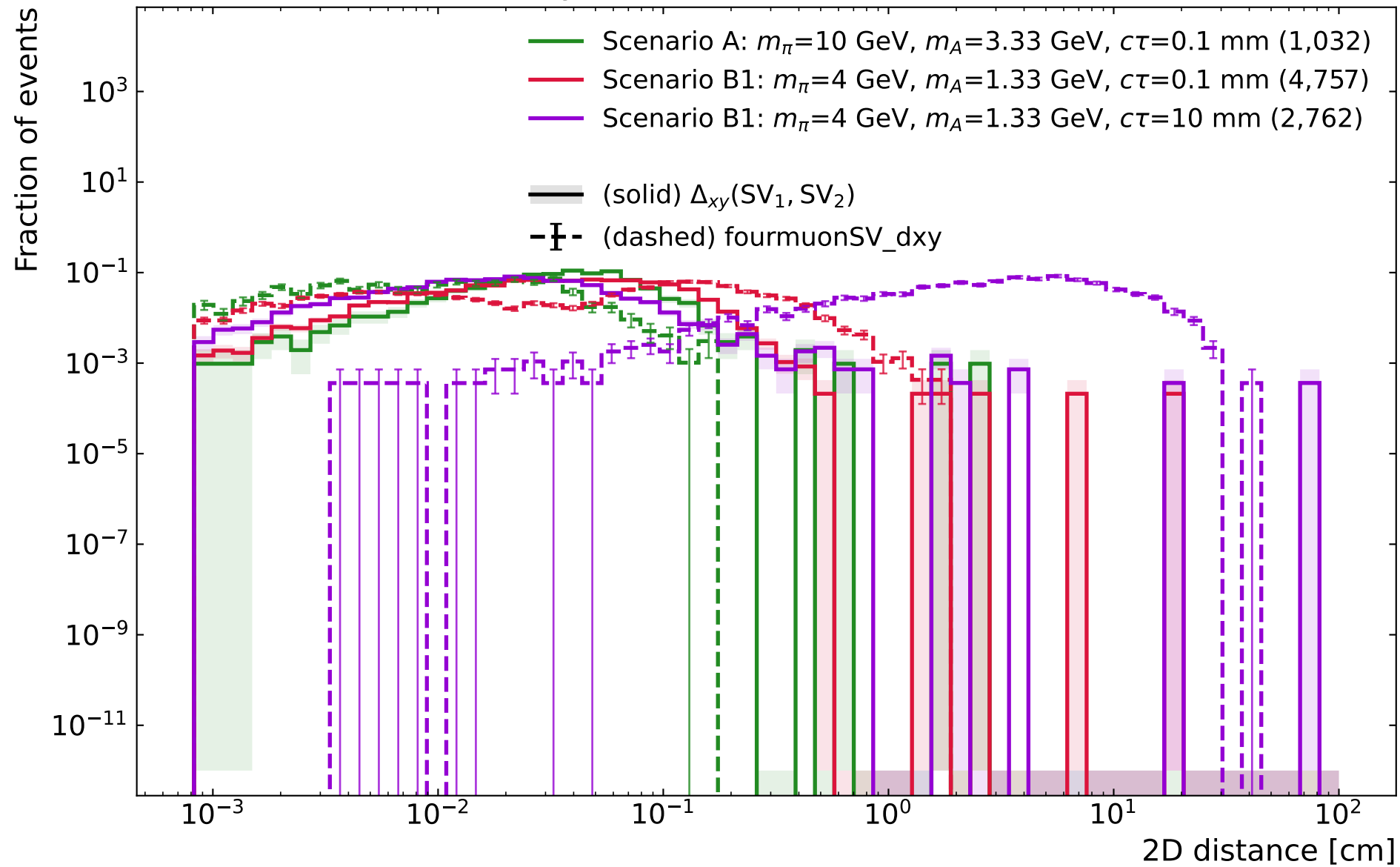
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
 trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
 each distribution is normalised to unit area; bands/bars are statistical only

CMS *Simulation Preliminary*

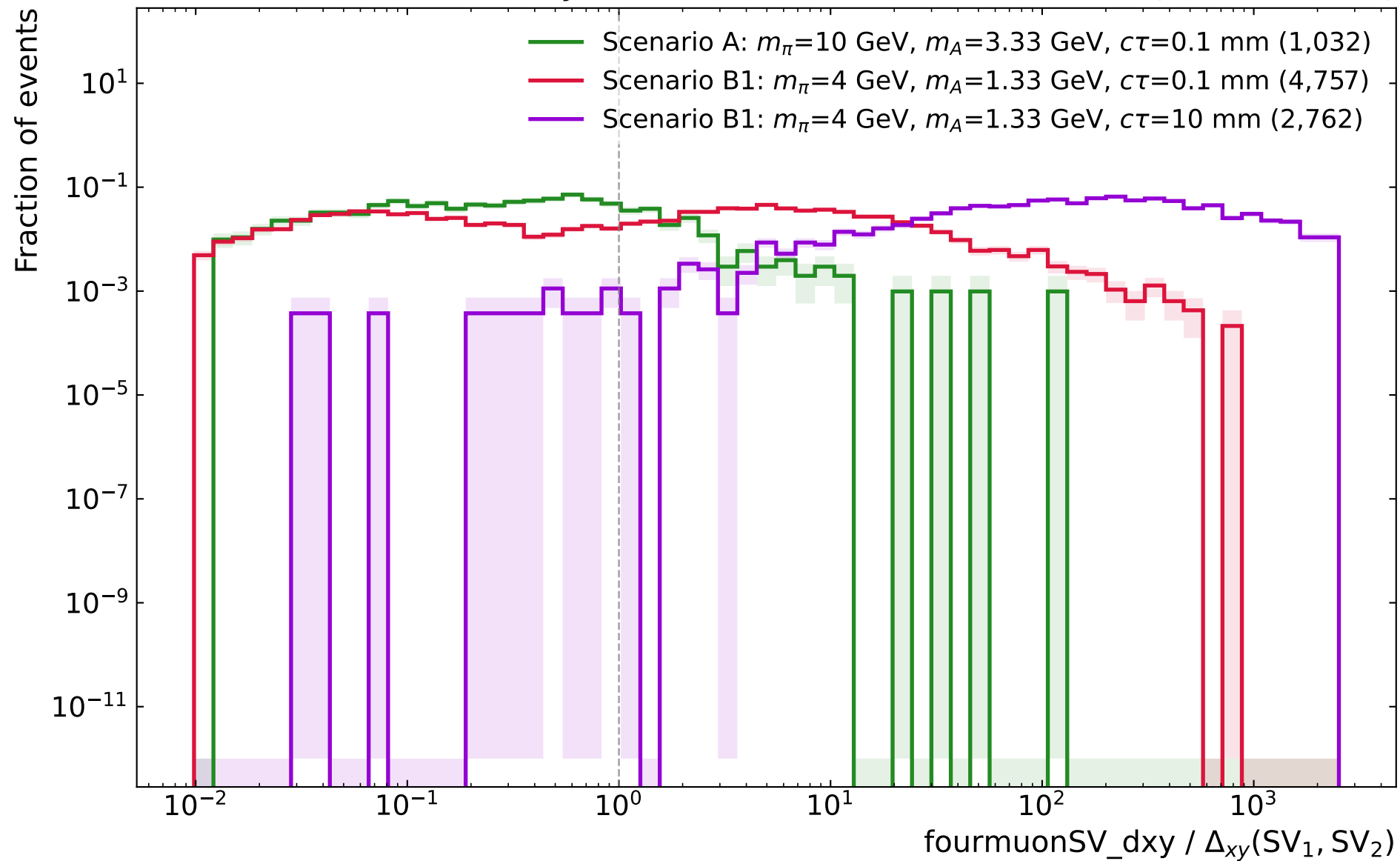
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
 trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
 each distribution is normalised to unit area; bands/bars are statistical only

CMS *Simulation Preliminary*

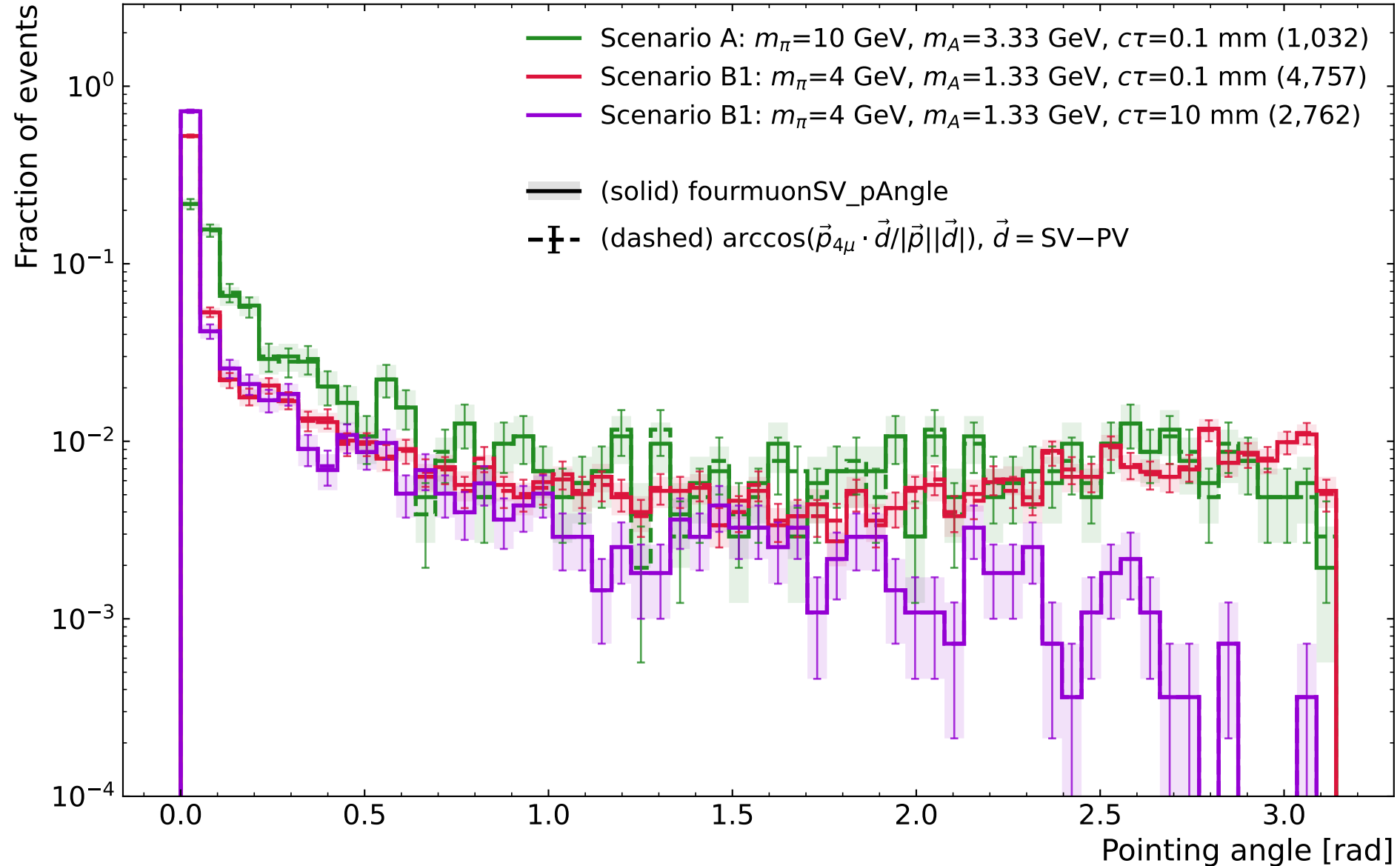
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
 trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
 each distribution is normalised to unit area; bands/bars are statistical only

CMS Simulation Preliminary

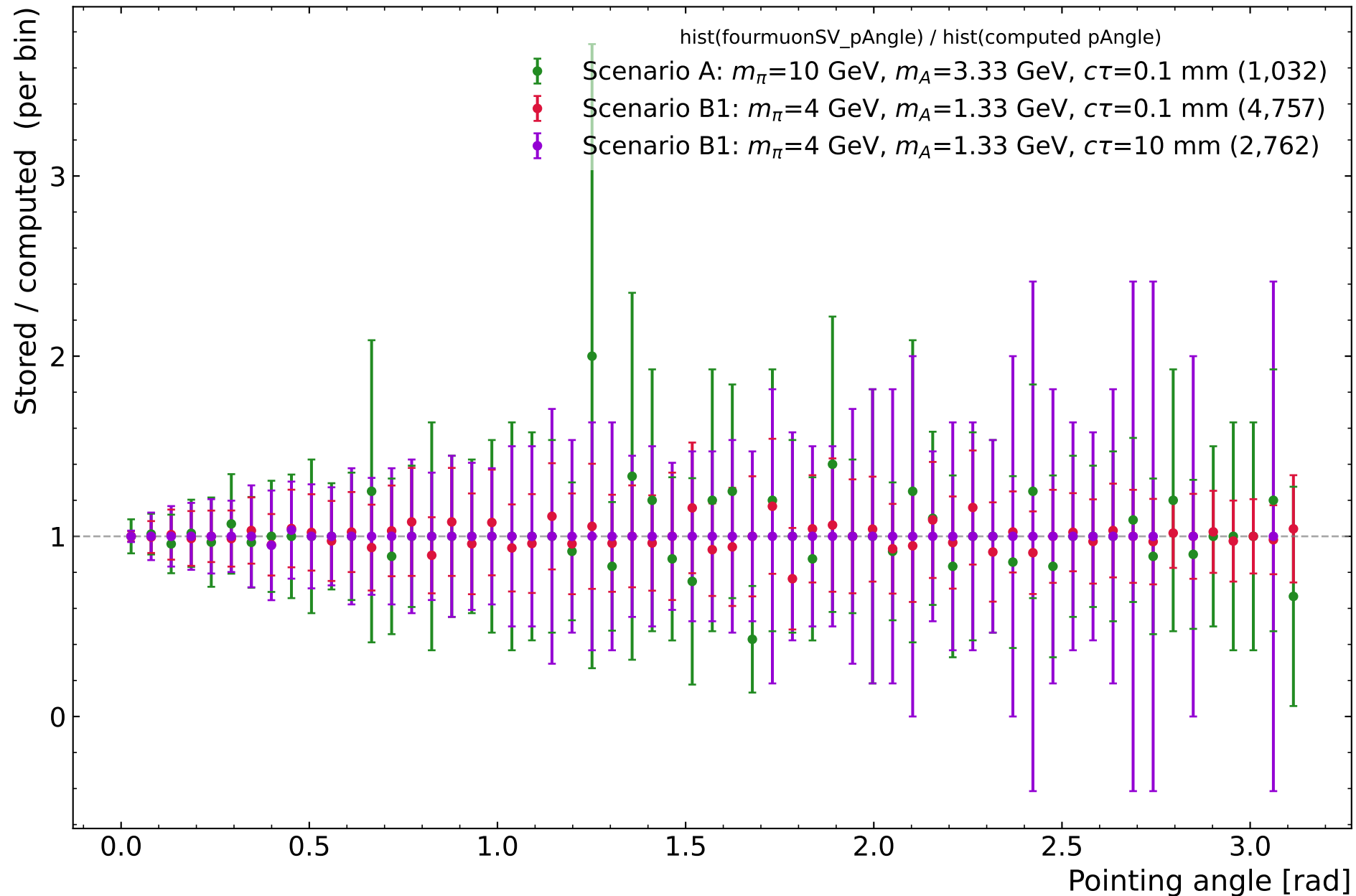
(13.6 TeV, 2024)



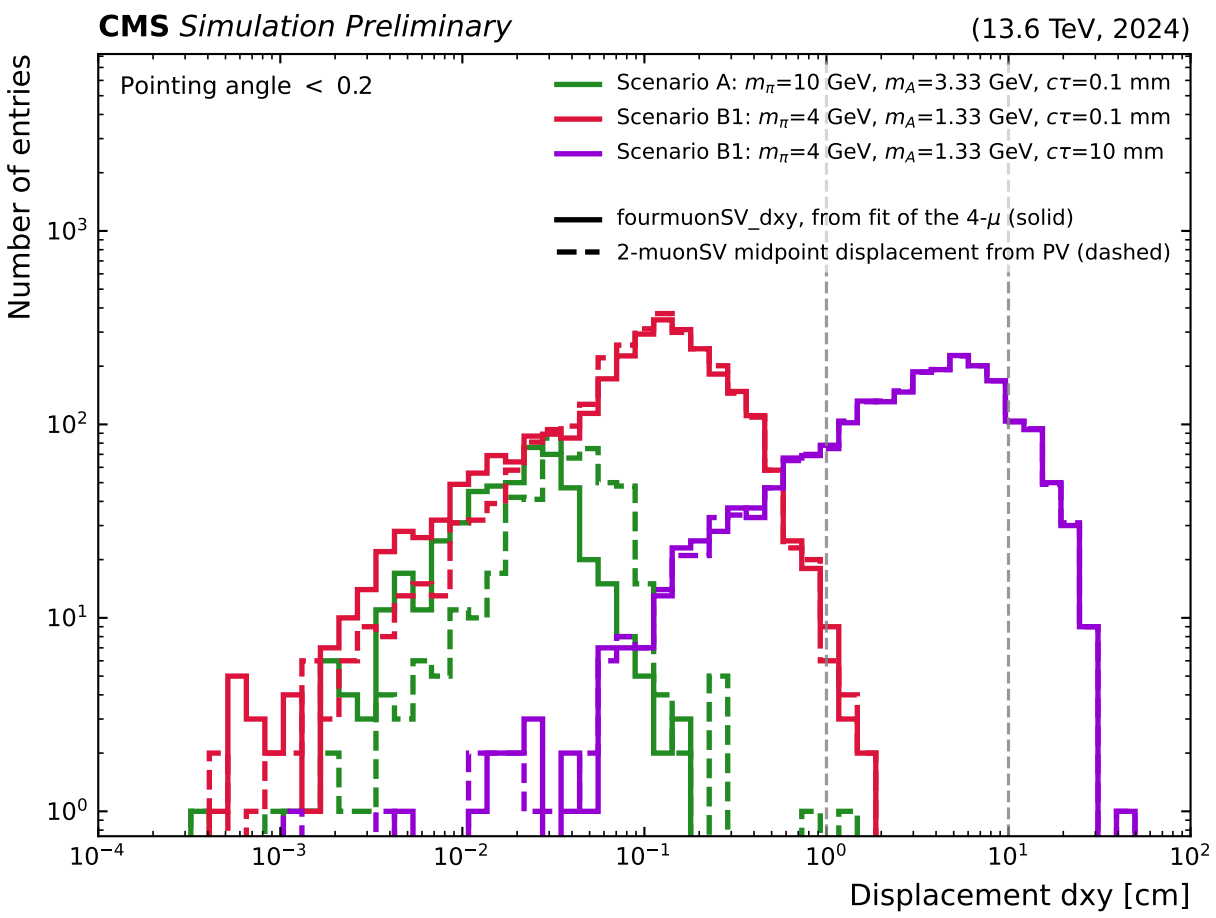
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection

CMS Simulation Preliminary

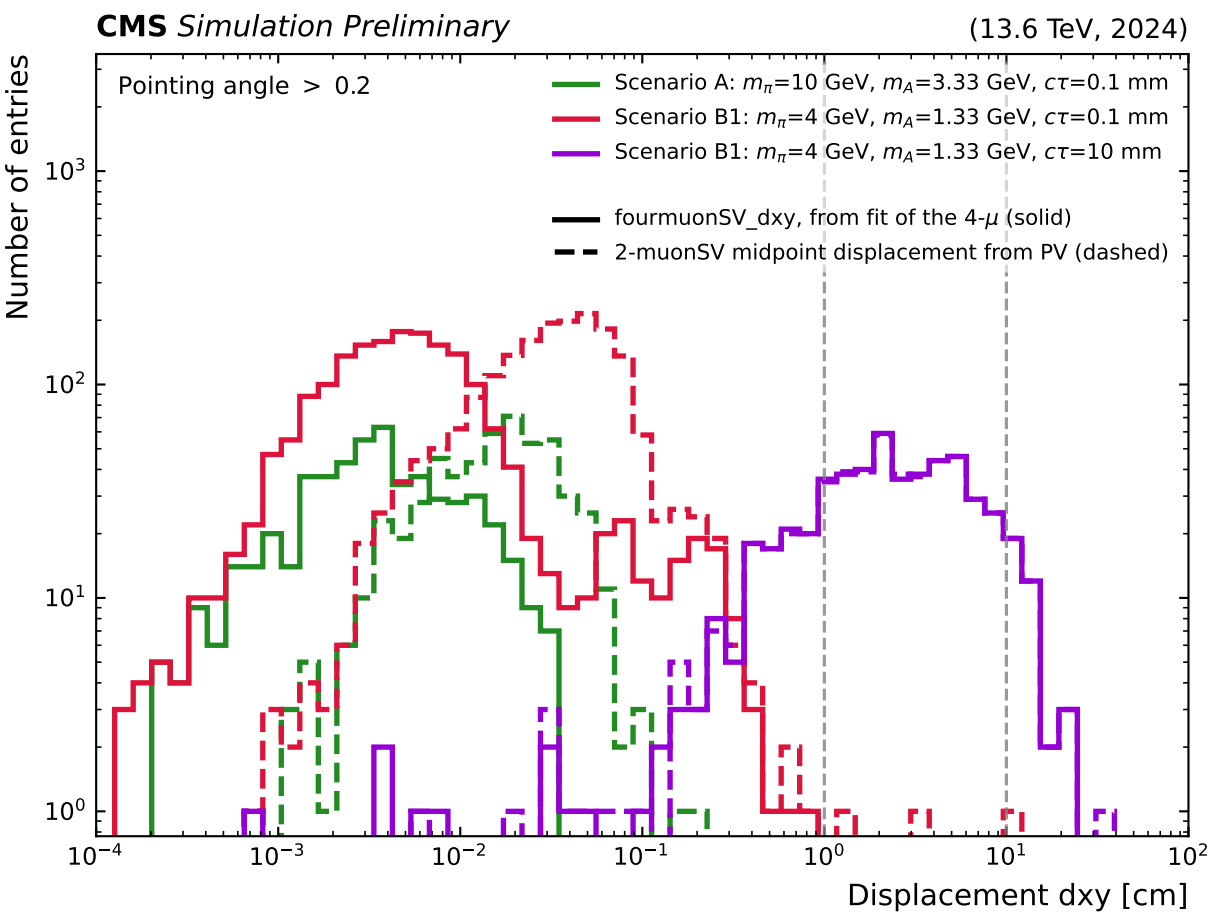
(13.6 TeV, 2024)



quadv category populations vs dxy, split by fourmuonSV_pAngle
solid: fourmuonSV_dxy, from fit of the 4- μ dashed: 2-muonSV midpoint displacement from PV grey lines: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	500	0	0	500
A $c\tau=0.1$ 2muSV mid	499	1	0	500
B1 $c\tau=0.1$ fourmuonSV	2,906	11	0	2,917
B1 $c\tau=0.1$ 2muSV mid	2,906	11	0	2,917
B1 $c\tau=10$ fourmuonSV	409	1,552	267	2,228
B1 $c\tau=10$ 2muSV mid	408	1,556	264	2,228



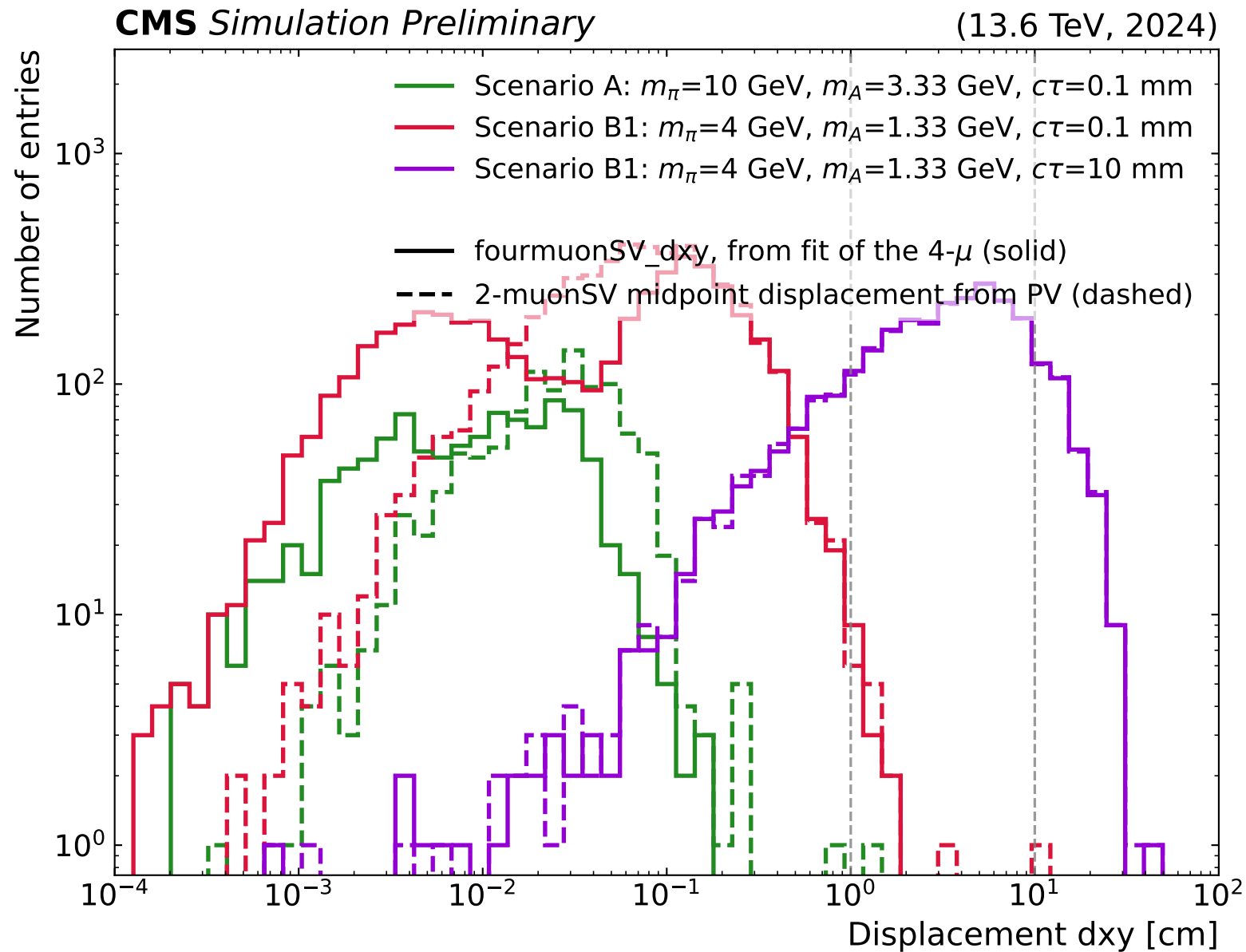
	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	532	0	0	532
A $c\tau=0.1$ 2muSV mid	532	0	0	532
B1 $c\tau=0.1$ fourmuonSV	1,840	0	0	1,840
B1 $c\tau=0.1$ 2muSV mid	1,837	3	0	1,840
B1 $c\tau=10$ fourmuonSV	118	386	30	534
B1 $c\tau=10$ 2muSV mid	117	385	32	534

quadv category populations vs dxy (pAngle-inclusive)

solid: fourmuonSV_dxy, from fit of the 4- μ

dashed: 2-muonSV midpoint displacement from PV

grey lines: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	1,032	0	0	1,032
A $c\tau=0.1$ 2muSV mid	1,031	1	0	1,032
B1 $c\tau=0.1$ fourmuonSV	4,746	11	0	4,757
B1 $c\tau=0.1$ 2muSV mid	4,743	14	0	4,757
B1 $c\tau=10$ fourmuonSV	527	1,938	297	2,762
B1 $c\tau=10$ 2muSV mid	525	1,941	296	2,762

Ratio of dxy distributions: fourmuonSV_dxy / (2-muonSV midpoint dxy)
grey dashed: dxy category edges (1, 10 cm); dotted: ratio = 1

